AGV For Material Handling

Modelling and simulation of dynamic systems

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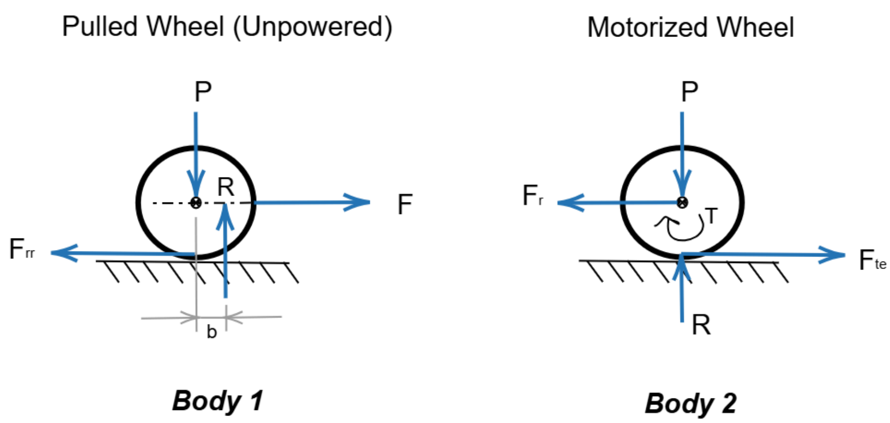
Table Of Contents

I. System Schematics & Equations Of Motion



Body 1: it is the caster wheel that is not powered by any motors , the forces acting on it are

- Rolling resistance force which is the opposition of the wheel to the rolling that results from the deformation of a wheel against the ground.
- the pulling force
- the reaction force , and due to the deformation happening in the wheel , the stress concentration is not uniform which will result that the reaction force will be displaced distance b from the center of mass of the wheel

Body 2: it is a motorized wheel that it is attached to the agv and the forces acting on it are :

- the load force that we are trying to pull
- the torque that is resulted from the motor
- the reaction force acting in the center of mass
- the tractive effort is the result of the static friction that occurs between the wheel and the ground as a result of a torque being applied to the wheel by a motor . This force is responsible for moving the vehicle forward and must be able to overcome

1. Force of Gravity on an Inclined Plane
2. Aerodynamic Drag
3. Rolling resistance

For Body 1

$$\sum F = m_{wheel} \times a$$

$$\boxed{F - F_{rr} = m_{wheel} \times a} \qquad [1]$$

$$\sum M_o = I \times \alpha$$

$$F_{rr} \times r - Rb = I \times \alpha$$

$$F_{rr} = \frac{1}{r}\,[I\alpha + Rb\,]$$

By Subistituting with

$$\alpha = \frac{a}{r}$$

$$\mu_{rr} = \frac{b}{r}$$

$$\boxed{F_{rr} = I\,\frac{a}{r^2} + R\,\mu_{rr}} \qquad [2]$$

The deformation of the tire against the floor is the rolling resistance . This deformation shifts the normal force (R) by a distance (b), which creates the rolling resistance moment that the tractive force Fte must continuously overcome.

where

$m_{wheel} \implies$ mass of the wheel
$a \implies$ acceleration
$r \implies$ radius of the wheel
$b \implies$ horizontal distance between R and the centre of the wheel
$P \implies$ vertical load on the wheel including weight
$R \implies$ normal reaction
$F \implies$ force pulling the wheel
$I \implies$ moment of inertia of the wheel
$\alpha \implies$ angular acceleration of the wheel
$\mu_{rr} \implies$ rolling friction coefficient

For Body 2

$$\sum F = m_{wheel} \times a$$

$$F_{te} - F_r = m_{wheel} \times a$$

$$\boxed{F_r = F_{te} - m_{wheel} \times a} \qquad [6]$$

$$\sum M_o = I \times \alpha$$

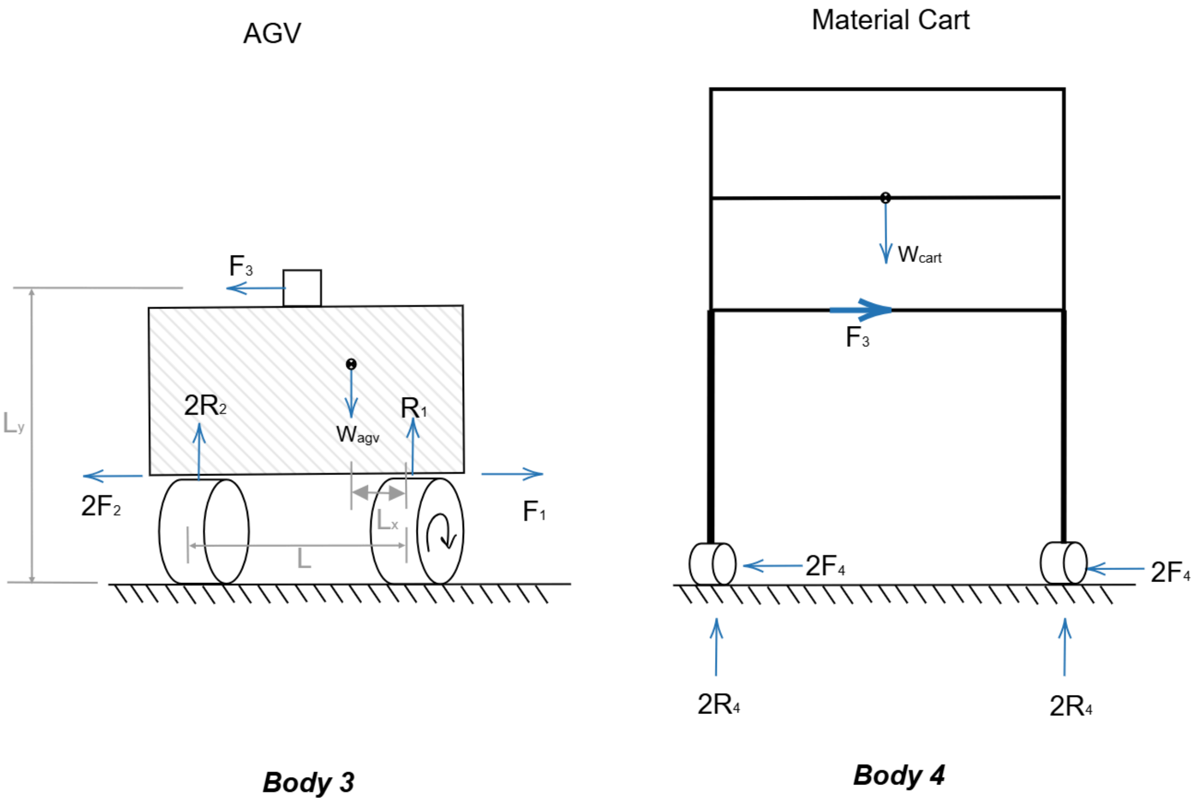
$$T - F_{te} \times r = I\,\alpha$$

By Subistituting with

$$\alpha = \frac{a}{r}$$

$$\boxed{T = I\frac{a}{r} + F_{te} \times r} \qquad [7]$$

For the drive wheel, friction is what pushes the AGV forward instead of slowing it down. The motor torque (T) must provide enough power to overcome the wheel's rotational weight and drag the trailing load (Fr), creating the forward push known as Tractive Effort (Fte)



Body 3: it is the automated guided vehicle (agv)with motorized wheels and a material cart that is loaded vertically on the agv , but the material cart is modelled as if the agv pulls the cart for simplification(i guess , i am not sure). And the forces acting on the agv are :
-F1 is the tractive force Fte
- F2 and F4 is the rolling resistance of each one of the wheels
- F3 is the force pulling the cart
- R1 and R2 are the normal reactions for the wheels
- the total weight of the agv at the center of mass
Body 4 : it is the material cart that consists of multiple stories all made as an external frame to cary in the material inside it , the forces acting on it is :
- F4 is the rolling resistance of each one of the wheels
-weight of the cart
- R4 is the normal reactions at the caster wheels

For Body 3

$$\sum F = m_{vehicle} \times a$$

$$F_{te} - F_{rr} - F_{ad} - F_{slope} - \sum F_{external,\,x} = m_{vehicle} \times a$$

$$\boxed{F_{te} = m_{vehicle} \times a + F_{rr} + F_{ad} \mp F_{slope} + \sum F_{external,\,x}} \qquad [5]$$

$$\boxed{F_{slope} = W \sin(\psi)} \qquad [4]$$

$$\boxed{F_{ad} = \frac{1}{2} \rho_{air} v^2 C_D A} \qquad [3]$$

$$\boxed{W_{agv} = m_{agv}\, g}$$

-Aerodynamic Drag Drag is the force that a fluid will exert on a moving body that is opposite to its relative motion. Aerodynamic drag (Fad) is the term used when the surrounding fluid is air
- Force of Gravity on an Inclined Plane When a vehicle is moving on a plane with inclination is called Fslope
For the case of this project,the aerodynamic drag and the slope were disregarded.This is because the ground is plane all throughout the workshop,so the AGV is not expected to encounter any slopes. Besides this, the AGV will not move at great speed, so the aerodynamic drag will be very small in comparison to the rest of the forces.

where

$F_{ad} \implies$ *Aerodynamic drag*

$\rho_{air} \implies$ *air density*

$C_D \implies$ *drag coefficient*

$v \implies$ *relative speedbetween fluid and body*

$A \implies$ *characteristic area*

$F_{slope} \implies$ *Force of Gravity on an Inclined Plane*

$W \implies$ *weight*

$\psi \implies$ *angle of the plane with inclination*

$F_{te} \implies$ *Tractive Effort*

$F_r \implies$ *load being pulled*

$F_{external,\,x} \implies$ *external forces acting on the vehicle in the direction ofmotion*

$T \implies$ *torque*

$m_{vehicle} \implies$ *mass of the vehicle*

From Equations $1, 2, 3, 4, 5, 6, 7$

$$
\begin{cases}
R_4 = \dfrac{W_{cart}}{4} \\[2ex]
F_4 = I_4 \dfrac{a}{r_4^2} + \mu_{rr4} R_4 \\[2ex]
F_3 = m_{cart}\, a + 4F_4 \\[2ex]
R_2 = \dfrac{F_3\, y + W_{agv}\, x}{2L} \\[2ex]
R_1 = W_{agv} - 2R_2 \\[2ex]
F_2 = I_2 \dfrac{a}{r_2^2} + \mu_{rr2} R_2 \\[2ex]
F_1 = m_{agv}\, a + 2F_2 + F_3 \\[2ex]
I_2 = I_4 = \dfrac{1}{2} m_{wheel}\, r_{wheel}^2
\end{cases}
\qquad [8]
$$

We did not include air resistance (aerodynamic drag) or ground slope. This is because the workshop floor is flat, so the AGV won't go up or down any hills. Also, the AGV will move slowly, meaning the air resistance force will be very small compared to other forces.
$F_1 \implies F_{te}$ $F_2 \implies F_{rr}$ $(rolling\ resistance\ of\ each\ individual\ pulled\ wheel)$ $F_3 \implies$ $force\ pulling\ the\ cart$ $F_4 \implies F_{rr}$ $(rolling\ resistance\ of\ each\ individual\ pulled\ wheel)$ $L_x \implies$ $the\ horizontal\ distance\ between\ front\ wheel\ and\ center\ of\ gravity\ of\ theAGV$ $L_y \implies$ $the\ vertical\ distance\ between\ F3\ and\ the\ ground$ $L \implies$ $the\ horizontal\ distance\ between\ front\ wheel\ and\ back\ wheels\ of\ the\ AGV$

II. Linerization

The equations do not contain any non-linear terms because air drag and slope equations were excluded. Therefore, the system does not need to be linearized, and the equations can be solved and simulated directly.

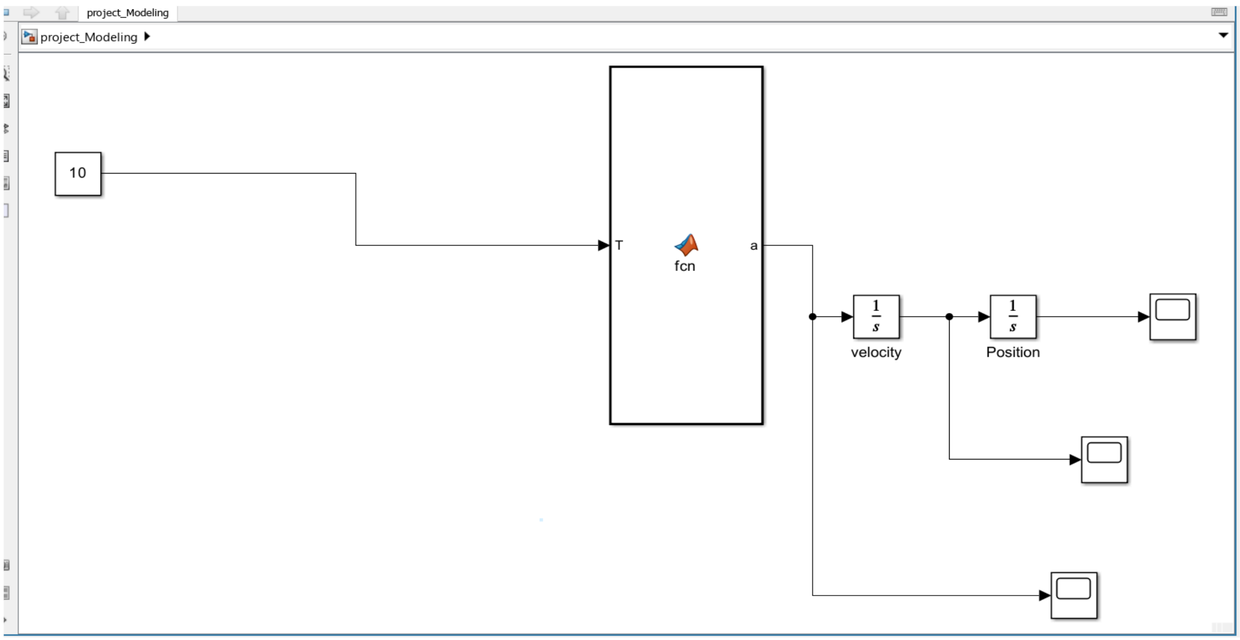
III. MATLAB implementation

project_M_and_S.m

C:\Users\mahmo\Documents\MATLAB\project_M_and_S.m

```
% 1. Universal Constants
g = 9.81; % Gravitational acceleration [m/s^2]

% 2. Masses and System Geometry
m_agv  = 50;   % Mass of the AGV unit [kg]
m_cart = 100;  % Mass of the Trailer/Cart [kg]
L      = 0.5;  % Wheelbase (Distance between front and rear axles) [m]
x      = 0.25; % Horizontal distance from front axle to Center of Gravity [m]
y      = 0.3;  % Vertical height of the towing/hitch point [m]

% 3. Wheel and Rotational Properties
r        = 0.1; % Wheel radius [m]
m_wheel = 2;   % Mass of a single wheel [kg]
% Mass Moment of Inertia for a solid cylindrical wheel (I = 0.5 * m * r^2)
I_w      = 0.5 * m_wheel * r^2;

% 4. Rolling Resistance Coefficients (Friction)
mu_2 = 0.015; % Coefficient of rolling resistance for AGV wheels
mu_4 = 0.02;  % Coefficient of rolling resistance for Cart wheels

% 5. Control Input
T = 10; % Applied Motor Torque per drive wheel [N.m]
```

MATlap Standard Inputs



Simulink- System Representation

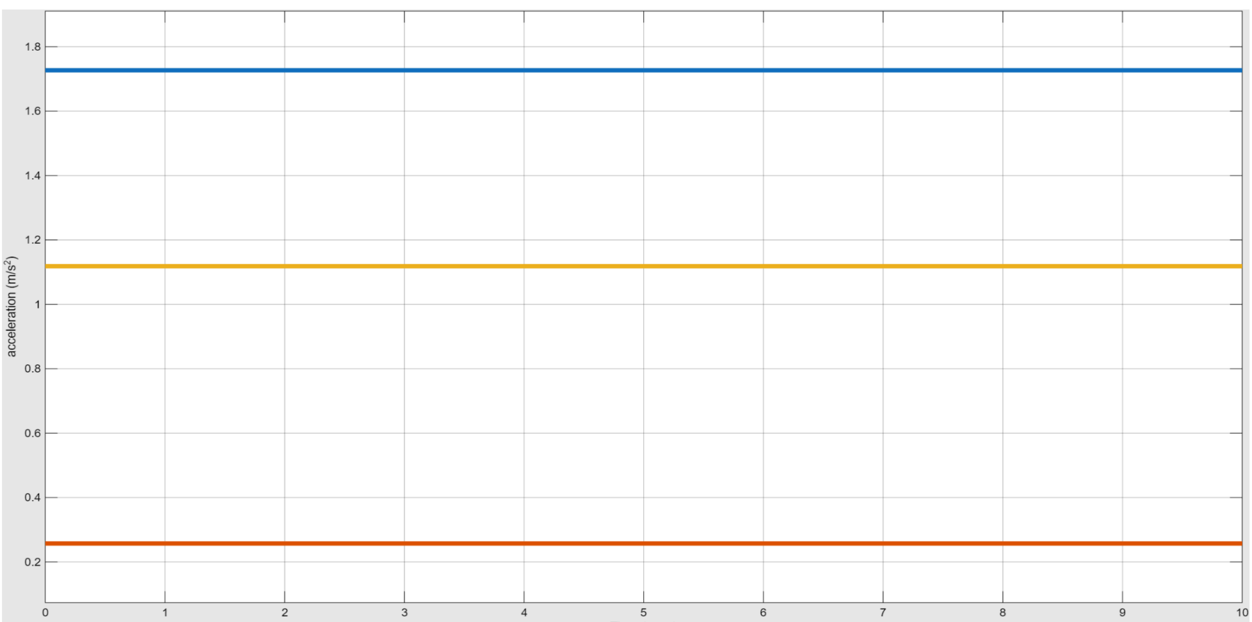
```
function a = fcn(T, m_agv, m_cart, I_w, mu_2, mu_4, x, g, r, L)



    W_agv = m_agv * g;
    W_cart = m_cart * g;

    R2 = (W_agv * x) / (2 * L);
    R4 = W_cart / 4;

    Denominator = m_agv + m_cart + (8 * I_w / (r^2));

    Numerator = (2 * T / r) - (2 * mu_2 * R2 + 4 * mu_4 * R4);

    a = Numerator / Denominator;
end
```

MATlab Function

In the MATlab representation of the system we assume the AGV robot and the car as one body and we are dealing with the total forces acting on the system including the driving forces and resisting forces .
So that , we consider F3 as internal force because it resist the motion of AGV and at the same time it is considered the driving force for the cart,according to Newton's second law we conclude the general equation of motion for the whole system :

a =((Total driving forces)-(Total resistance))/Total mass .

$T = F * d = F * r \Rightarrow F = T/r$. (driving force of one motor)
Because we have two motors for the AGV So, the total driving force is equal (2T / r).
We also ignore the friction of the two motorized wheels because it is very small and does not affect the motion .
Total resistance = 2 *u_{rr2} *R2 (two pulled wheels of AGV) + 4 * u_{rr4} * R4 (4 wheels of the cart).
Total mass (effective mass) = m_agv + m_cart +8* (2 * I_{w} / r^2) . We use the moment of inertia to get the mass of the wheels because it affects the spinning of wheels.



Case Study 1: The Effect of Mass on Acceleration.

We notice that the acceleration is constant because we applied a constant torque from the motors on the cart and we ignored the air resistance .

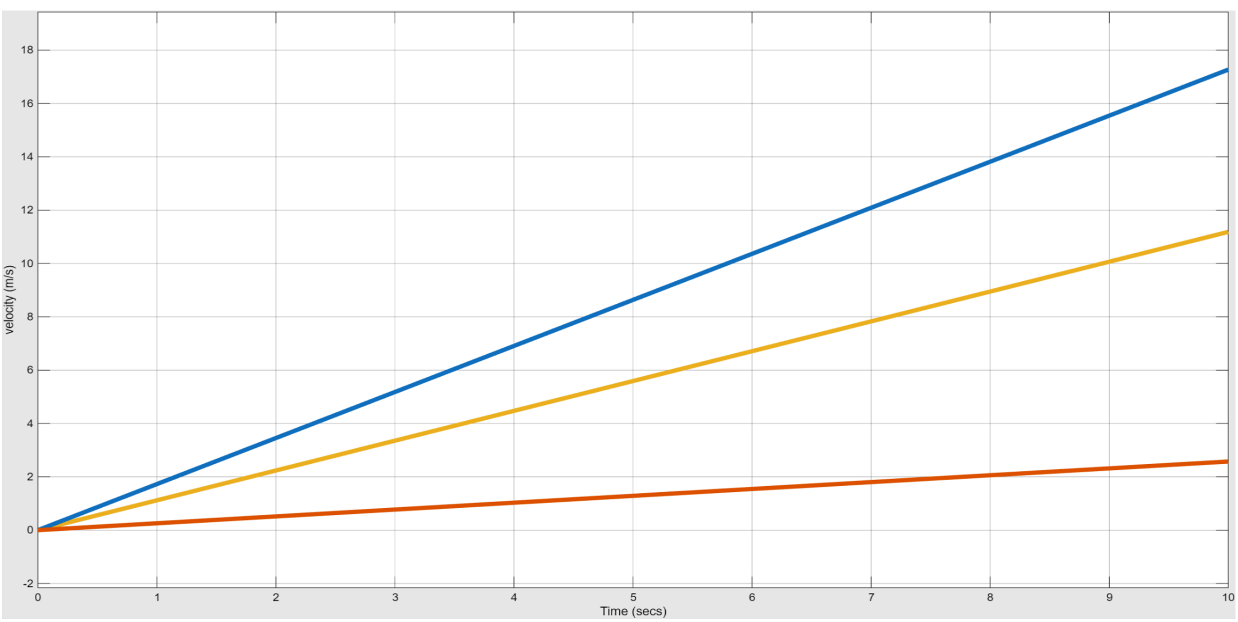
Yellow line : we use mass of cart(m_cart =100 kg) ,as we see on the graph the acceleration (a ~ = 1.15 m/s^2) .

Blue line : we use mass of cart(m_cart =50 kg) ,as we see on the graph the acceleration has increased (a ~ = 1.75 m/s^2) .

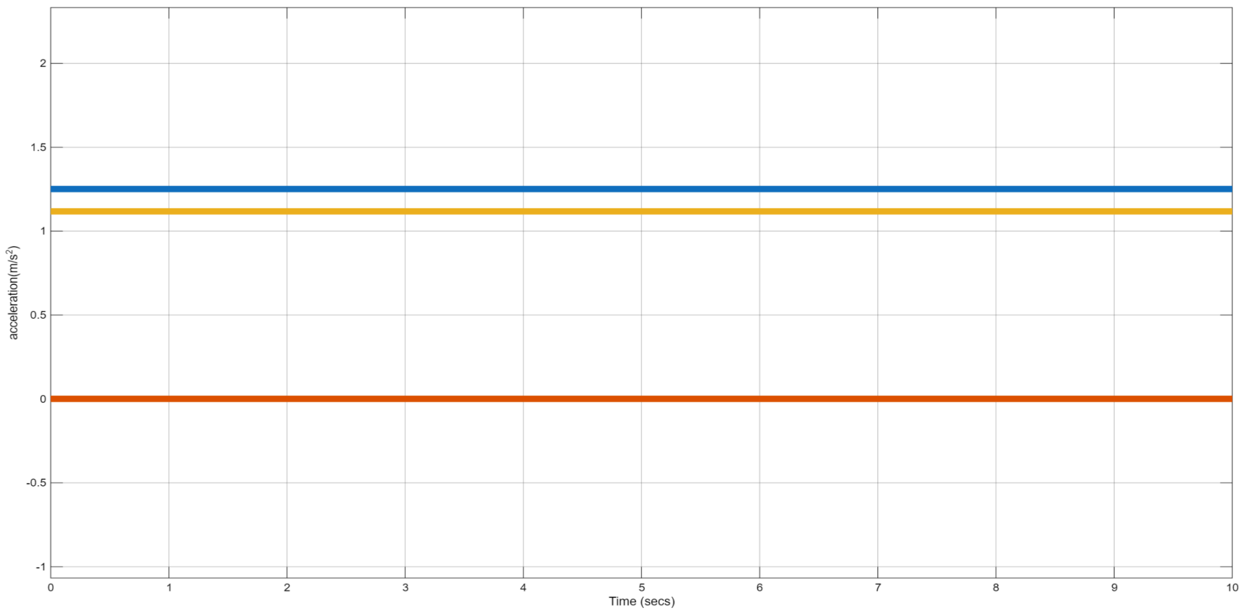
Red line : we use mass of cart(m_cart =400 kg) ,as we see on the graph the acceleration has decreased (a ~ = 0.27 m/s^2) .

We conclude that ,the Acceleration is inversely proportional with Mass of cart and the above graph is considered the prove for the relation:

(a =((Total driving forces)-(Total resistance))/Total mass).



*And by logic the velocity will increase with the small mass and will decrease with the large mass similar to the acceleration due to the relation (dv/dt =a).



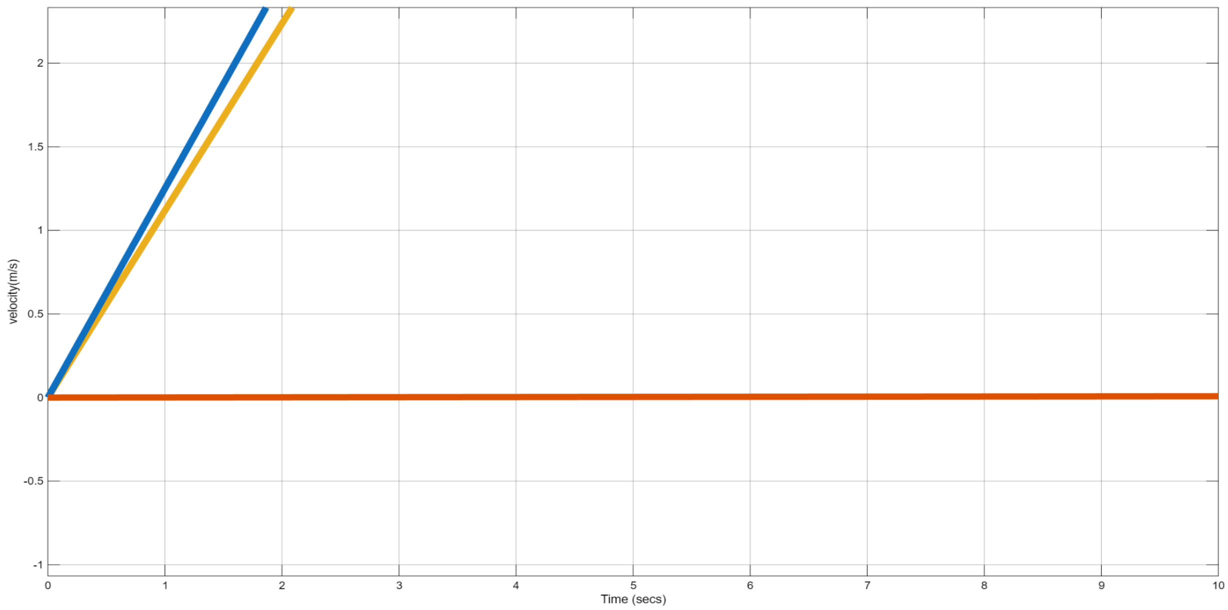
Case Study 2: The Effect of Rolling Friction on Acceleration.

Yellow line : we use rolling friction coefficients (u_rr2 = 0.015 and u_rr4 =0.02) ,as we see on the graph the acceleration (a ~ = 1.15 m/s^2) .

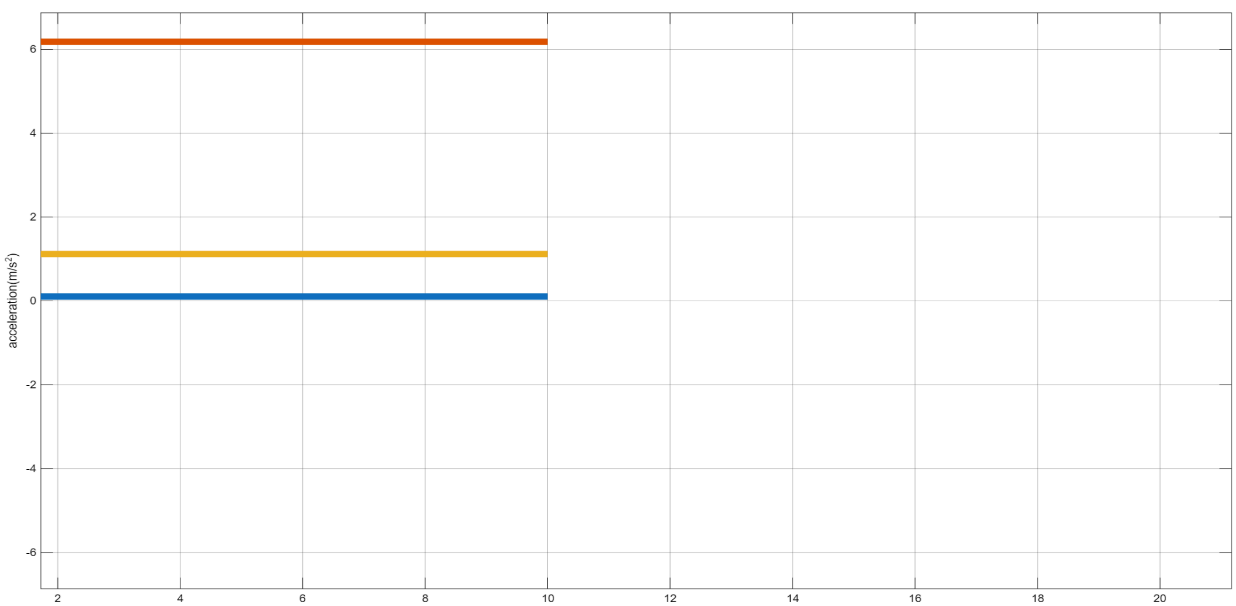
Blue line : we use rolling friction coefficients (u_rr2 = 0.0015 and u_rr4 =0.002) ,as we see on the graph the acceleration there is a small increase of the acceleration (a ~ = 1.3m/s^2) .

Red line : we use the critical rolling friction coefficients (u_rr2 =0.163 and u_rr4 =0.163) ,as we see on the graph the acceleration is equal to zero (a = 0 .00m/s^2) this mean system at equilibrium .

We notice that , there are a inversely relation between the acceleration and the rolling friction coefficient ,



*And here as we see the velocity equal to zero when we use the critical rolling friction coefficient , the system is in equilibrium so,the cart and AGV are at rest , and there is a small increase in the velocity while decreasing the rolling friction coefficient.



Case Study 3: The Effect of Torque on Acceleration.

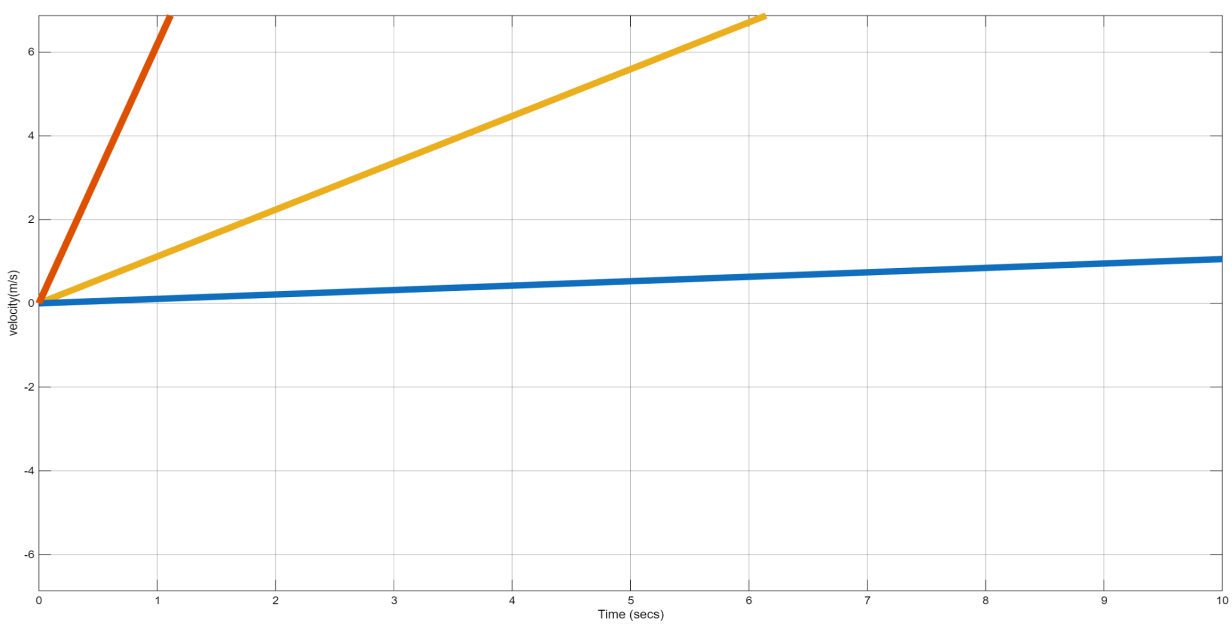
Yellow line : we use torque (T = 10 N\m) ,as we see on the graph the acceleration (a ~ = 1.15

m/s^2) .

Blue line : we use torque (T = 2 N\m) ,as we see on the graph the acceleration is very small approximately equal to zero (a ~ = 0.2 m/s^2)

Red line : we use torque (T = 50 N\m) ,as we see on the graph the acceleration increased (a ~ = 6.15 m/s^2) .

We notice that the acceleration is proportional to the torque .



* Here is the increase in velocity due to the increase of the acceleration while increasing the torque

Conclusion

The acceleration is directly proportional to the torque and inversely proportional to the rolling friction coefficient and the mass .